

Soft Robotic Jumping Glider

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ABSTRACT

This paper presents the design, development, and testing of a novel soft robotic platform capable of vertical lift-off, inspired by biological systems. The system integrates a spring-actuated jumping mechanism, with the long-term goal of enabling gliding using deployable wings. The primary objective is to enable lightweight aerial mobility for reconnaissance, environmental monitoring, and exploration missions. The robot, weighing approximately 70g, employs carbon fiber spring elements actuated by a motor-latch trigger system for vertical lift-off. At present, the platform has achieved a 10-foot jump in final prototype tests. Although wing deployment was conceptually designed, it was not implemented in the current prototype and is planned for future iterations.

Keywords: Soft robotics, jumping glider, bio-inspired mobility, spring actuator, aerial robot

1. INTRODUCTION

Modern robotics has increasingly drawn inspiration from biology to develop energy-efficient and adaptive systems capable of navigating complex terrains. One such innovation is the soft robotic jumping glider, a hybrid system that emulates the biomechanical strategies of flying animals by combining vertical takeoff via spring actuation and post-lift gliding. Existing drone platforms suffer from high energy demands and limited terrain adaptability. Our approach seeks to overcome these limitations through a novel lightweight mechanism that minimizes energy consumption while maximizing range and versatility.

While jumping robots have demonstrated high agility in terrestrial locomotion [1], and gliding mechanisms have been explored independently [2], combining these in a compact, soft robotic form offers distinct advantages. Our work builds on spring actuation concepts [3] and deployable wing technologies [4], adapting them into a system suitable for aerial deployment, terrain mapping, and planetary exploration.

2. METHOD

2.1. Design Overview

The proposed robotic platform is engineered to achieve vertical lift-off through a motor-actuated, spring-loaded mechanism. At the core of the design is a high-tension energy storage system employing carbon fiber strips, functioning as resilient springs due to their high modulus-to-weight ratio. The energy storage is controlled by a precision-g geared DC motor that incrementally winds the carbon fiber strips via a pulley system. A custom-designed latch mechanism, fabricated using laser-cut acrylic and integrated with a metal pin-and-slot interface, ensures energy retention until the precise moment of launch. The mechanism is configured for minimal energy loss and optimized for rapid energy discharge to enable vertical acceleration.

The design incorporates interface brackets for passive wing deployment to complement future gliding capability. These brackets include anchor points for torsional springs and are dimensioned based on load simulations to ensure the wings deploy under aerodynamic pressure or triggered actuation.

2.2. Structural and Material Considerations

The structural design prioritizes lightweight durability and impact resistance. The main chassis is constructed from 4 mm carbon fiber rods held in a truss arrangement for load distribution. The body enclosure is 3D-printed using Acrylonitrile Butadiene Styrene (ABS) with a 40% infill to balance strength and mass. Stress analysis was performed using ANSYS Static Structural simulations to validate that the spring release force does not deform the housing beyond the elastic limit.

Carbon fiber strips used for spring actuation are composed of unidirectional prepreg layups with an approximate elastic modulus of 70 GPa. The latching assembly utilizes stainless steel fasteners for high fatigue resistance and consistent engagement under cyclic loading. The robot's mass is distributed symmetrically along its central axis to avoid instability during launch. A fixed ballast is used to tune the center of gravity, ensuring minimal pitching during upward motion.

2.3. Prototype Development

Prototype Development Prototype development followed a three-stage iterative process, incorporating both mechanical and electronic subsystems:

- **Iteration 1 (Mechanical Validation):** A non-actuated physical model was built to test the structural alignment and basic spring dynamics. Manual winding was used to validate energy storage and mechanical robustness.
- **Iteration 2 (Integrated Subsystem Development):** A brushed DC gear motor (torque: 1.2 kg·cm @ 6V) and an Arduino Nano controller were added. The electrical harness was routed internally, and custom PCBs were fabricated to manage motor control, latch actuation, and telemetry.
- **Iteration 3 (Full System and Field Testing):** The third iteration involved full-scale outdoor tests. Improvements included aerodynamic shaping of the body, protective landing skid integration, and the implementation of a repeatable release mechanism using a micro-servo (SG90). The motor wind-up sequence was optimized through encoder feedback to avoid over-compression of the spring.



Image 2.1

2.4. Test Procedure

To evaluate performance under realistic conditions, the robot was tested in an outdoor environment with low wind interference (<2 m/s). A high-speed camera (240 fps) placed orthogonally to the launch trajectory captured the vertical motion. A metric reference board was positioned in the background to calibrate the frame-based height measurements. Each trial involved charging the spring to full tension, releasing the latch, and recording the trajectory.

Key performance metrics included:

- **Vertical Jump Height:** Measured across 10 trials, with an average height of 9.75 feet ($\sigma = \pm 0.22$ ft).
- **Stability Metrics:** Pitch and yaw angles were derived from onboard IMU logs and video frame overlays.
- **Repeatability:** The standard deviation in vertical height and lateral drift provided insight into the system's consistency.
- **Energy Transfer Efficiency:** Estimated by comparing the input motor torque and observed kinetic energy at take-off.

Failure modes such as premature latch release, asymmetric spring deflection, or off-axis thrust were monitored and documented. Insights from the test cycles informed subsequent design adjustments, including latch strengthening and CG tuning through ballast modification.

2.5. Electronics and Control Systems

The electronics suite includes an Arduino Nano microcontroller, a brushed DC motor driver (TB6612FNG), a 9-DOF IMU (MPU-9250), and a barometric pressure sensor (BMP388). The firmware implements finite state machines for charging, locking, and launching sequences. Power is supplied through a 3.7V 500mAh LiPo cell, with protection circuitry. Serial telemetry allows ground-based monitoring of jump events and motor diagnostics.

2.6. Prototyping and Testing Pipeline



Image 2.2

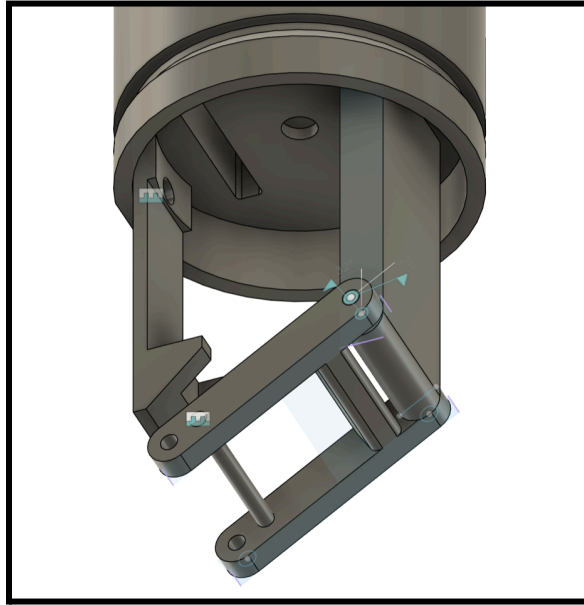


Image 2.3

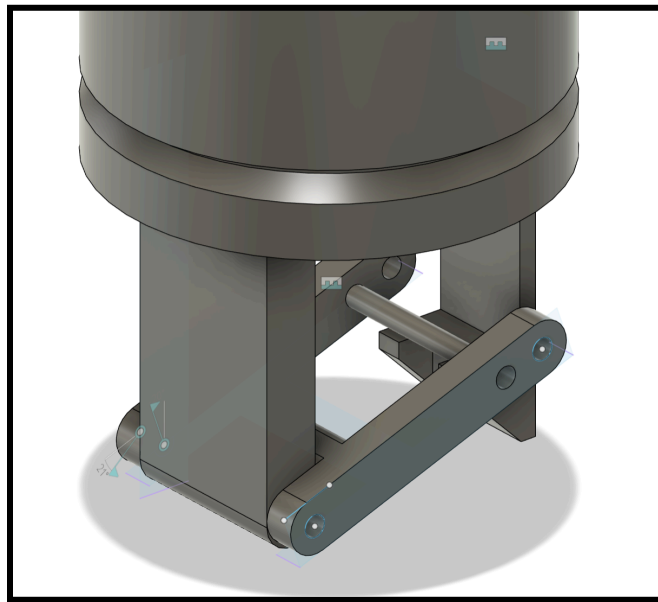


Image 2.4

- Iteration 1: 3D-printed body frame with manual spring compression.
- Iteration 2: Integration of DC motor-driven winding system and mechanical latch.
- Iteration 3: Full integration of electronics with jump trials and performance logging.

2.7. Experimental Setup Testing

The experiment was conducted on a flat open field. High-speed video capture (240 fps) measured vertical displacement and angular deviation during launch. Ten trials were conducted, and height data were processed via frame analysis. Stability was assessed based on yaw/pitch drift during ascent.

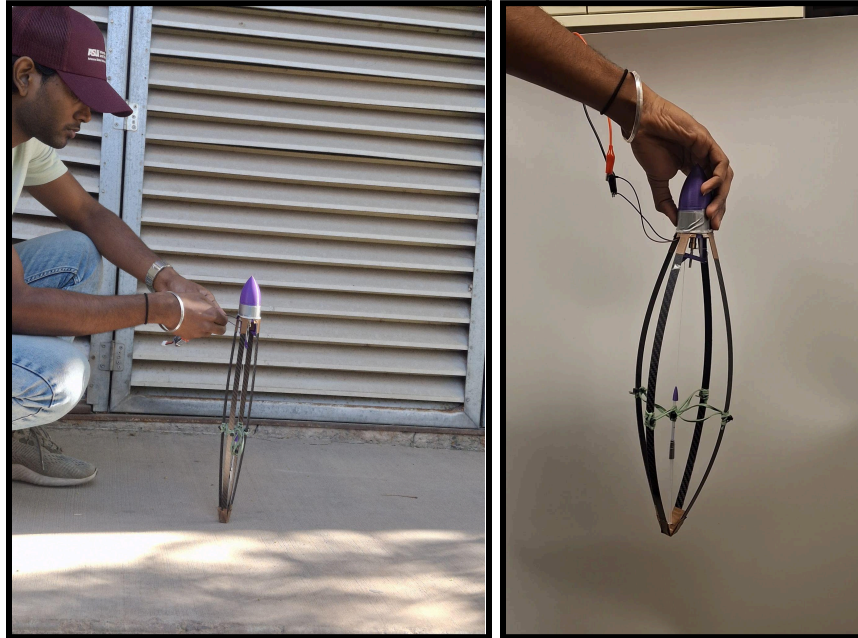


Image:-2.5,2.6

3. RESULTS AND DISCUSSION

3.1. Vertical Jumping Performance

The robot achieved a maximum jump height of approximately 10 feet using the spring-powered mechanism. The lift-off was clean and repeatable across multiple tests. The motor-based winding system successfully stored sufficient elastic energy without system failure.

3.2. Challenges and Limitations

Challenges included:

- Precision in motor control for energy storage.
- Latching mechanism reliability.

- Structural integrity after repeated tests.

3.3. Future Work

- Design and integration of deployable wings for controlled gliding.
- Implement IMU-based autonomous stabilization.
- Increase jump height to 60–80 feet through optimized spring design.
- Integrate camera payload for terrain mapping.
- Develop swarm operation capability.

4. CONCLUSION

This study demonstrated the feasibility of a lightweight, soft robotic jumping platform capable of vertical lift-off. Initial results validate the mechanical design and actuation concepts. The prototype achieved a 10-foot lift-off using spring energy, marking a significant milestone. Although gliding was not implemented, the conceptual design has been laid out, and future versions will aim to integrate and validate controlled aerial descent. The platform holds promise for applications in exploration, surveillance, and disaster response where traditional aerial systems may not perform optimally.

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